

Postdoc, Harvard University
tzuhan@seas.harvard.edu
(515) 441-6303
tzuhancs.github.io
tzuhanhsu
122 Citations

Tzu-Han Hsu



RESEARCH INTERESTS

My research focuses on developing algorithms for **verification** and **synthesis** using **formal methods** within the relational reasoning framework named **hyperproperties**, including applications in **privacy-aware** policies, **secure information-flow**, **reinforcement learning**, and **safety-guaranteed** analysis. My current work explores **integrating AI and large language models (LLMs) with formal methods**, bringing together the reasoning power of modern AI and the rigor of formal guarantees.

EDUCATION

Ph.D. in Computer Science and Engineering 2020 – 2026

Michigan State University, East Lansing, MI

GPA: 3.93/4.0 **Advisor:** Dr. Borzoo Bonakdarpour

🏆 2026 Three Minute Thesis (3MT) Competition Winner, MSU College of Engineering

🏆 2024 Best Research Poster Award, MSU College of Engineering

M.S. in Computer Science and Engineering

Michigan State University, East Lansing, MI

GPA: 3.93/4.0

🏆 2021 Google Research Fellowship Nomination, MSU College of Engineering

B.S. in Computer Science 2016 – 2020

Iowa State University, Ames, IA

GPA: 3.86/4.0

🏆 Graduated *Magna Cum Laude*

🏆 2013 – 2020 ISU College of Liberal Arts and Sciences Dean's List

B.M. in Piano Performance 2013 – 2017

Iowa State University, Ames, IA

GPA: 3.86/4.0

🏆 Outstanding Senior Nomination, ISU Department of Music

RESEARCH EXPERIENCE

Postdoctoral Research Fellow @ Harvard University 2026 – Present

Advisor Prof. Nada Amin

Topic Integrating AI and Large Language Models (LLMs) with Formal Methods

Graduate Research Assistant @ Michigan State University

2020 – 2026

Advisor Prof. Borzoo Bonakdarpour

Topics REASONING ABOUT HYPERPROPERTIES THROUGH SYMBOLIC DECISION PROCEDURES

Keywords: HyperLTL, SMT Solving, Symbolic Execution, SyGuS, Runtime Enforcement, Program Repair, Policy Synthesis, Model Checking, Reinforcement Learning

Research Intern @ Microsoft Research (MSR)

Summer 2023

Mentors Shuo Chen, Nikolaj Bjørner

Project THEOREM-CARRYING-TRANSACTION: PROVABLE SECURITY FOR SMART CONTRACTS

Keywords: Solidity, Boogie, EVM, Go-Ethereum, ERC20, UniSwap-V2

Applied Scientist Intern @ Amazon AWS Automated Reasoning Group

Summer 2022

Mentors Aman Goel, Michael Whalen

Project SYMBOLIC MODEL CHECKING FOR DISTRIBUTED PROTOCOLS WITH PARALLELIZATION

Keywords: P Language, Symbolic Execution, AWS Lambda, Amazon S3

Undergraduate Research Assistant @ Iowa State University

2019 – 2020

Advisor Prof. Borzoo Bonakdarpour

Project BOUNDED MODEL CHECKING FOR HYPERPROPERTIES

Keywords: QBF/SMT Solvers, Z3, QuAbS, Linear Temporal Logic, Information-Flow Security

JOURNAL PAPERS

- [j1] Tzu-Han Hsu, Ana Oliveira da Costa, Andrew Wintenberg, Ezio Bartocci, Borzoo Bonakdarpour
“**Gray-box Runtime Enforcement of Hyperproperties**”
Acta Informatica (2025).

CONFERENCE PAPERS

- [c1] Tzu-Han Hsu, Milad Rabizadeh, Kenneth Rogale, Fedor Filippov, Marco A. de Oliveira Batista, and Borzoo Bonakdarpour
“**HyperQB2.0: A Bounded Model Checker for Hyperproperties**”
International Conference on Computer Aided Verification (CAV’26), to appear.
- [c2] Arshia Rafieioskouei, Tzu-Han Hsu, Matthew Lucas, and Borzoo Bonakdarpour
“**HyPOLE: Hyperproperty-Guided Multi-Agent Reinforcement Learning under Partial Observation**”
International Conference on Machine Learning (ICML’26), to appear.
- [c3] Tzu-Han Hsu, Arshia Rafieioskouei, Borzoo Bonakdarpour
“**HyprL: Reinforcement Learning of Control Policies for Hyperproperties**”
Annual Conference on Neural Information Processing Systems (NeurIPS’25), San Diego, CA.
- [c4] Raven Beutner, Tzu-Han Hsu, Borzoo Bonakdarpour, and Bernd Finkbeiner
“**Syntax-Guided Automated Program Repair For Hyperproperties**”
International Conference on Computer Aided Verification (CAV’24), Montreal, Canada.
- [c5] Tzu-Han Hsu, Borzoo Bonakdarpour, Bernd Finkbeiner, and César Sánchez
“**Bounded Model Checking for Asynchronous Hyperproperties**”
International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS’23), Paris, France.
- [c6] Tzu-Han Hsu, César Sánchez, Sarai Sheinvald, and Borzoo Bonakdarpour
“**Efficient Loop Conditions for Bounded Model Checking Hyperproperties**”

International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS'23), Paris, France.

[c7] Tzu-Han Hsu, Borzoo Bonakdarpour, Eunsuk Kang, and Stavros Tripakis
“**Mapping Synthesis for Hyperproperties**”
IEEE International Symposium on Computer Security Foundations (CSF'22/FLoC'22), Haifa, Israel.

[c8] Tzu-Han Hsu, César Sánchez, and Borzoo Bonakdarpour
“**Bounded Model Checking for Hyperproperties**”
International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS'21), Luxembourg, Luxembourg.

ARXIV PREPRINTS

- [a1] Thomas Ball, Nikolaj S. Bjørner, Ashley J. Chen, Shuo Chen, Yang Chen, Zhongxin Guo, Tzu-Han Hsu, Peng Liu, Nanqing Luo
“**Theorem-Carrying Transactions: Runtime Verification to Ensure Interface Specifications for Smart Contract Safety (v2)**”
- [a2] Tzu-Han Hsu, Yu Wang, Borzoo Bonakdarpour, Miroslav Pajic
“**Multi-Agent Path Planning with Hyperproperties (v1)**”

RESEARCH FELLOWSHIPS

- **Michigan State University Graduate Research Fellowship** 2021, 2022
Received Summer Support for outstanding students selected by MSU CSE department.
- **Google Research Fellowship Nomination** 2021
Selected as one of the only four nominees selected by MSU College of Engineering.

RESEARCH AWARDS

- **Three Minute Thesis (3MT) Competition Winner** 2026
Awarded by the Michigan State University College of Engineering.
- **Best Research Poster Award** 2024
Awarded by Michigan State University College of Engineering Graduate Research Symposium.

INVITED RESEARCH TALKS

- **Purdue University Programming Languages Group (PurPL) Seminar** 2025
Hosted by prof. Benjamin J Delaware
- **Academia Sinica (National Academy of Scientific Research, Taiwan)** 2023
Hosted by prof. Yu-Fang Chen
- **Portland State University Department of Computer Science PLV Seminar** 2022
Hosted by prof. Yao Li
- **University of California Santa Cruz: Languages, Systems, and Data Seminar** 2021
Hosted by prof. Lindsey Kuper and prof. Tyler Sorensen
- **RWTH Aachen University, Germany** 2021
Hosted by prof. Joost-pieter katoen and prof. Erika Ábrahám

PROFESSIONAL SERVICES

· Program Committee:

OOPSLA	<i>ACM SIGPLAN Conf. on OOP, Systems, Languages & Applications</i>	'27
EMSOFT	<i>International Conference on Embedded Software</i>	'26
LMPL	<i>ACM SIGPLAN Workshop on Language Models and Programming Languages</i>	'26
EMSOFT	<i>International Conference on Embedded Software, WiP-LB TPC</i>	'25

· Conference Paper Reviews:

CSL	<i>European Association for Computer Science Logic</i>	'25
SRDS	<i>International Symposium on Reliable Distributed Systems</i>	'24
EMSOFT	<i>International Conference on Embedded Software</i>	'24
CAV	<i>International Conference on Computer Aided Verification</i>	'24
RV	<i>International Conference on Runtime Verification</i>	'24,'25
TACAS	<i>Tools and Algorithms for the Construction and Analysis of Systems</i>	'23,'25
FM	<i>International Symposium on Formal Methods</i>	'23
ICCPS	<i>International Conference on Cyber-Physical Systems</i>	'23
VSTTE	<i>Verified Software: Theories, Tools, Experiments</i>	'23
NFM	<i>NASA Formal Methods</i>	'23
DISC	<i>International Symposium on Distributed Computing</i>	'22
CSF	<i>Computer Security Foundations Symposium</i>	'22
ATVA	<i>Int'l Symposium on Automated Technology for Verification and Analysis</i>	'21

· Conference Artifact Evaluation:

CAV	<i>International Conference on Computer Aided Verification</i>	'23,'24
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· Journal Reviews:

ACTA	<i>Acta Informatica</i>	'22,'25,'26
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TEACHING EXPERIENCE

- **Graduate Teaching Assistant** Spring 2020
Michigan State University, CSE 260: Discrete Structures in Computer Science
- **Undergrad Teaching Assistant** Spring, Fall 2018
Iowa State University, COM S 227: Introduction to Object-Oriented Programming

HONORS AND SCHOLARSHIPS

- **Outstanding Senior Nomination** by ISU Music Department 2020
- **Dean's List** of Iowa State University College of Liberal Arts and Sciences 2013–20
- **Future Leader Award** issued by U.S. Ambassador to China Terry Branstad 2018
- **Gold Level Scholarship Recipient** as Iowa State University International Student Ambassador 2013–18
- **Outstanding Student Scholarship Recipient** by Iowa State University Music Department 2013–18

LEADERSHIP AND VOLUNTEER EXPERIENCES

- **Research Conference Student Volunteer** at Computer Aided Verification for information publicity 2021

- **Certificate of Achievement** from The Society for Collegiate Leadership & Achievement 2016–20
- **Department Student Representative** of Iowa State University Student Council 2017–18
- **Volunteer/Event Coordinator** at Iowa Education without Borders (non-profit organization) 2017–18
- **Student Mentor** at Iowa Youth Leadership Summer Camp 2017–18
- **Student Volunteer** at Iowa State University Annual K-12 Computational Thinking Competition 2020

MUSIC AWARDS AND HONORS

- **Assistant Concertmaster (Violin)** at Iowa State University Symphony Orchestra 2015-17
- **Soloist Winner - Piano Division** Fort Dodge Orchestra Young Artists Concerto Competition 2015
- **1st Prize Winner** Iowa State University Symphony Orchestra Concerto Competition 2014
- **State-Level Competition Winner** Music Teacher National Association (MTNA) Piano Competition 2014

INDUSTRIAL EXPERIENCE

Software Engineer Intern 2019–20

Worrell Medical Design Inc., Minneapolis, MN

- Developed a Virtual Reality Environment on *Oculus Quest* for usability testing in Human Factors Engineering.
- Programmed well-functioning C# scripts in *Unity3D* engine with desired behaviors and outcomes.
- Analyzed user feedback and debugged/adjusted/improved the existing versions of the program with human factors engineers.